Experiment No. : 1

Experiment Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Experiment Program Code

Code 1: InHead.html (Internal JS in Head Section)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>JavaScript in Head</title>
    <script>
        function myFunction() {
            document.getElementById("heading2").innerHTML = "Heading changed.";
        }
    </script>
</head>
<body>
    <h2 id="heading2">Demo JavaScript in Head</h2>
    <p id="demo">A paragraph.</p>
    <button type="button" onclick="myFunction()">Try it!</button>
</body>
</html>
```

Code 2: InBodySection.html (Internal JS in Head Section)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=<device-width>, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h2>Javascript in Body</h2>
    <p id = "demo"></p>

    <script>
        document.getElementById("demo").innerHTML = "My First JavaScript Program.";
    </script>
</body>
</html>
```

Code 3: Windows1.html (window alert())

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script>
        window.alert(5 + 6);
```

```
    </script>
</body>
</html>
```

Code 4: WindowsPrint.html (window.print())

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h2>The window.print() Method</h2>
    <p>Click the button to print the current page.</p>
    <button onclick="window.print()">Print Page</button>
</body>
</html>
```

Code 5: DocumentsWrite.html (document.write())

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=<device-width>, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h1>My First Web Page</h1>
    <p>My first paragraph</p>
    <script>
        document.write(5 + 6);
    </script>
</body>
</html>
```

Code 6: INLINE_JS.html (Inline JavaScript)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Inline JS Demo</title>
</head>
<body>
    <h2>Inline JavaScript Example</h2>
    <button onclick="alert('Welcome to JavaScript !!')">Click Me !!</button>
</body>
</html>
```

Code 7: INTERNAL_JS.html (Internal Javascript)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=<device-width>, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h2>Internal JavaScript Example</h2>
    <p>This is a paragraph.</p>
    <script>
        document.write(5 + 6);
    </script>
</body>
</html>
```

Code 8a: EXTERNAL_JS.html (External JS : HTML file)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>External JS Demo</title>
</head>
<body>
    <h2>External Javascript Demo</h2>
    <p id="demo">Welcome !!</p>
    <button onclick="changeText()">Click Me !!</button>
    <script src="script.js"></script>
</body>
</html>
```

Code 8b: script.js (External JS : JavaScript file)

```
function changeText() {
    document.getElementById("demo").innerHTML = "Hello ! This text is generated by javascript";
}
```

Code 9: Console.html (console.log)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Console Log Demo</title>
</head>
<body>
    <script>
        console.log(5 + 6);
    </script>
</body>
</html>
```

Code 10: ConsoleTable.html (console.table)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Console Table Demo</title>
</head>
<body>
    <h2>Javascript console.table() Demo</h2>
    <script>
        let students = [
            { RollNo: 201, Name: "Rahul", Course: "BTech" },
            { RollNo: 202, Name: "Priya", Course: "MTech" },
            { RollNo: 203, Name: "Amit", Course: "BArch" }
        ];
        console.table(students);
    </script>
</body>
</html>
```

Code 11: ConsoleTrace.html (console.trace)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Console Trace Demo</title>
</head>
<body>
    <button onclick="startProgram()">Click Here !!</button>
    <script>
        function startProgram() {
```

```
            displayMessage();
        }
        function displayMessage() {
            console.log("Welcome to JavaScript !!");
            console.trace("Function Call Trace");
        }
    </script>
</body>
</html>
```

4. Output

Code 1: InHead.html (Internal JS in Head Section)



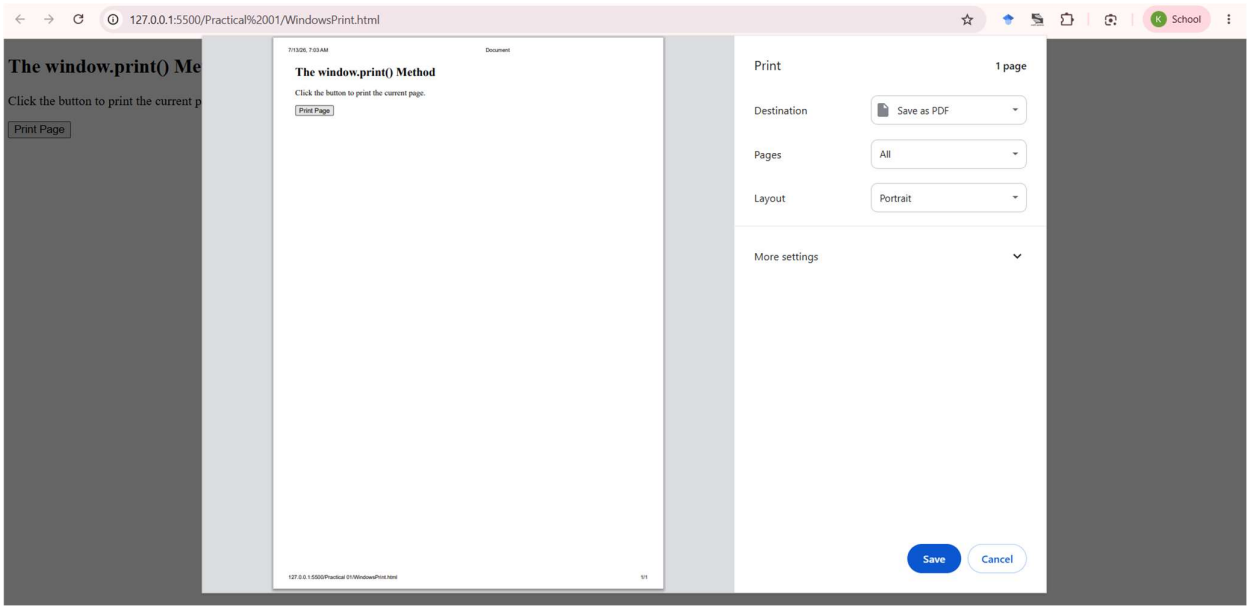
Code 2: InBodySection.html (Internal JS in Head Section)



Code 3: Windows1.html (window alert())



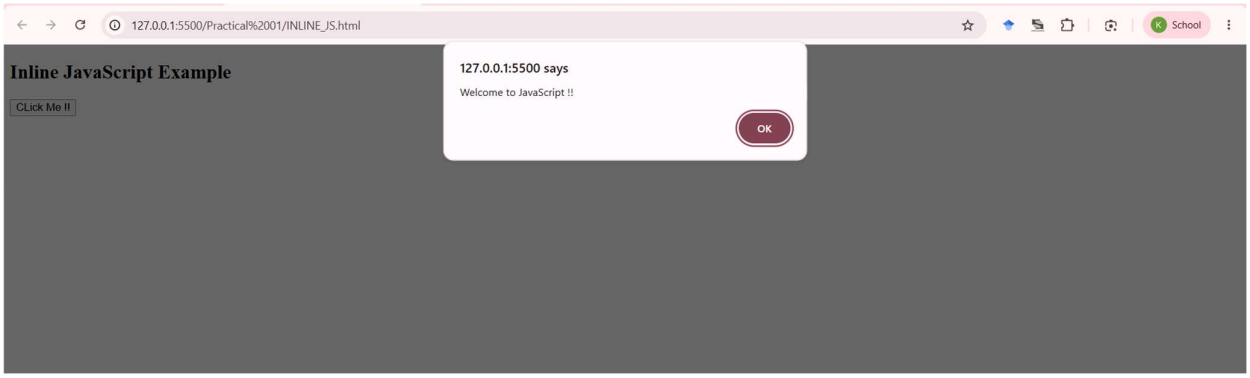
Code 4: WindowsPrint.html (window.print())



Code 5: DocumentsWrite.html (document.write())



Code 6: INLINE_JS.html (Inline JavaScript)



Code 7: INTERNAL_JS.html (Internal Javascript)



Code 8: EXTERNAL_JS.html (External JS : HTML file) and script.js (Javascript file)



Code 9: Console.html (console.log)



Code 10: ConsoleTable.html (console.table)



Code 11: ConsoleTrace.html (console.trace)



5. Case Study Title

SIT Nagpur Institute Webpage : Home Page Case Study (Using Inline, Internal JavaScript, window.alert() and DOM Styling)

6. Case Study Program Code

Casestudy_01_Shweta_Kushwaha_1001.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>PRN: 24070521001 | Name: Shweta Kushwaha | File: c:\Users\Shweta\Desktop\Symbiosis\Sem
5\JavaScript\Practical 01\Casestudy_01.html</title>
<style>
    :root {
        --red: #c62828;
        --red-dark: #8c0000;
        --blue: #113c5f;
        --light: #f7f7f7;
        --text: #333;
    }

    * { box-sizing: border-box; margin: 0; padding: 0; }
    body { font-family: Arial, sans-serif; color: var(--text); background: white; line-height: 1.6; }
    img { max-width: 100%; display: block; }
    a { text-decoration: none; color: inherit; }
    ul { list-style: none; }
    .container { width: 90%; max-width: 1100px; margin: auto; }

    .topbar { background: var(--red-dark); color: white; padding: 8px 0; font-size: 13px; }
    .topbar .container { display: flex; justify-content: space-between; flex-wrap: wrap; gap: 8px; }

    .navbar { background: white; padding: 12px 0; border-bottom: 3px solid var(--red); }
    .navbar .container { display: flex; justify-content: space-between; align-items: center; }
    .logo { display: flex; align-items: center; gap: 10px; }
    .logo-badge { background: var(--red); color: white; padding: 10px; border-radius: 50%; font-weight:
bold; }
    .logo-text h1 { font-size: 18px; color: var(--blue); }
    .logo-text span { font-size: 12px; color: var(--red); }

    nav ul { display: flex; gap: 20px; }
    nav a { color: var(--blue); font-weight: bold; }
    nav a:hover { color: var(--red); }

    .btn { display: inline-block; padding: 10px 16px; background: var(--red); color: white; border-radius:
5px; }
    .btn:hover { background: var(--red-dark); }

    .hero {
        background: linear-gradient(90deg, rgba(140, 0, 0, 0.95), rgba(17, 60, 95, 0.75)), url('campus.jpg')
center/cover;
        color: white;
```

```
        padding: 70px 0;
        text-align: center;
    }
    .hero h1 { font-size: 24px; margin-bottom: 10px; letter-spacing: 1px; }
    .hero h2 { font-size: 32px; margin-bottom: 15px; }
    .hero p { margin-bottom: 15px; }

    .image-grid { display: flex; gap: 20px; flex-wrap: wrap; margin-top: 30px; }
    .image-grid div { flex: 1 1 300px; }
    .image-grid img { width: 100%; border-radius: 10px; box-shadow: 0 6px 16px rgba(0,0,0,0.18); }

    .stats { display: flex; flex-wrap: wrap; gap: 15px; margin-top: -25px; position: relative; }
    .stats div { flex: 1; min-width: 160px; background: white; padding: 20px; border: 1px solid #eee; box-
shadow: 0 2px 8px rgba(0,0,0,0.08); text-align: center; }
    .stats h3 { color: var(--red); }

    .section { padding: 60px 0; }
    .about-grid { display: flex; gap: 30px; align-items: center; flex-wrap: wrap; }
    .img-block {
        flex: 1; min-width: 280px; padding: 120px 20px;
        background: linear-gradient(90deg, rgba(140, 0, 0, 0.75), rgba(17, 60, 95, 0.55)), url('about.jpg')
center/cover;
        color: white; border-radius: 10px;
    }
    .text-block { flex: 1; min-width: 280px; }

    .card-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(220px, 1fr)); gap: 20px; }
    .card { background: var(--light); padding: 20px; border-top: 4px solid var(--red); border-radius: 8px; }
    .card .icon { background: var(--blue); color: white; width: 42px; height: 42px; display: flex; align-
items: center; justify-content: center; margin-bottom: 10px; border-radius: 5px; }

    .why-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(200px, 1fr)); gap: 20px; }
    .why-item { background: var(--light); padding: 20px; border-left: 4px solid var(--red); }

    .cta { background: var(--blue); color: white; padding: 35px 20px; text-align: center; margin-top: 30px;
border-radius: 10px; }

    footer { background: #111; color: #ddd; padding: 40px 0 20px; margin-top: 40px; }
    .footer-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(220px, 1fr)); gap: 20px; }

    .nav-toggle { display: none; flex-direction: column; gap: 4px; cursor: pointer; }
    .nav-toggle span { width: 24px; height: 3px; background: var(--blue); }

    @media (max-width: 780px) {
        nav ul { display: none; }
        .nav-toggle { display: flex; }
        .navbar .container { position: relative; }
        nav ul {
            position: absolute;
            top: 60px;
            right: 0;
            background: white;
            flex-direction: column;
            padding: 15px;
            border: 1px solid #ddd;
        }
    }
</style>
</head>
<body>

    <div class="topbar">
        <div class="container">
            <span>Wathoda, Nagpur - 440008</span>
            <span>Constituent of Symbiosis International (Deemed University), Pune</span>
        </div>
    </div>

    <header class="navbar">
        <div class="container">
            <div class="logo">
                <div class="logo-badge">SIT</div>
                <div class="logo-text">
                    <h1>SIT NAGPUR</h1>
                    <span>SYMBIOSIS INSTITUTE OF TECHNOLOGY</span>
                </div>
            </div>
```

```
            <nav>
                <ul>
                    <li><a href="#about">About</a></li>
                    <li><a href="#academics">Academics</a></li>
                    <li><a href="#why">Why SIT</a></li>
                    <li><a href="#contact">Contact</a></li>
                </ul>
            </nav>
            <a href="#contact" class="btn btn-orange" style="display:none;" id="applyBtn">Apply Now</a>
            <div class="nav-toggle" id="navToggle"><span></span><span></span><span></span></div>
        </div>
    </header>

    <section class="hero">
        <div class="container">
            <h1>Symbiosis Institute of Technology, Nagpur</h1>
            <p class="eyebrow">ENGINEERING • RESEARCH • INNOVATION</p>
            <h2>Shaping Future-Ready Engineers Through Experiential Learning</h2>
            <p class="tagline">A constituent of Symbiosis International (Deemed University), Pune :
nurturing innovation, research and holistic growth on the principle of Vasudhaiva Kutumbakam.</p>
            <div class="btn-row">
                <a href="#academics" class="btn btn-orange">Explore Programs</a>
                <a href="#about" class="btn btn-outline-light">About the Institute</a>
            </div>
            <div class="image-grid">
                <div><img src="campus.jpg" alt="SIT Nagpur campus"></div>
                <div><img src="about.jpg" alt="SIT Nagpur campus building"></div>
            </div>
        </div>
    </section>

    <section class="section page-two">
        <div class="container">
            <div class="about-grid">
                <div class="text-block">
                    <h3>About Symbiosis Institute of Technology</h3>
                    <p>Symbiosis Institute of Technology, Nagpur is one of the leading engineering colleges
in Nagpur, nurtured by the visionary leadership of Symbiosis International (Deemed University), Pune.</p>
                    <p>The institute follows a Problem-Based Learning approach, encouraging students to
solve real-world challenges through hands-on projects, hackathons and industry-aligned coursework, supported
by state-of-the-art labs and a fully residential campus.</p>
                    <p class="highlight">Admissions open for industry-ready B.Tech programs with a strong
focus on engineering and research.</p>
                </div>
                <div class="image-block">
                    <img src="campus.jpg" alt="SIT Nagpur campus">
                </div>
            </div>
        </div>
    </section>

    <footer>
        <div class="container">
            <div class="footer-grid">
                <div>
                    <h4>SIT Nagpur</h4>
                    <p>Symbiosis Institute of Technology, Wathoda, Nagpur - 440008, Maharashtra, India</p>
                </div>
                <div>
                    <h4>Contact</h4>
                    <li>Phone: +91 90750 89800</li>
                    <li>Email: info@sitnagpur.siu.edu.in</li>
                </div>
            </div>
            <div class="footer-bottom">
                &copy; 2026 Symbiosis Institute of Technology, Nagpur. Demo page created as a JavaScript
case study project.
            </div>
        </div>
    </footer>

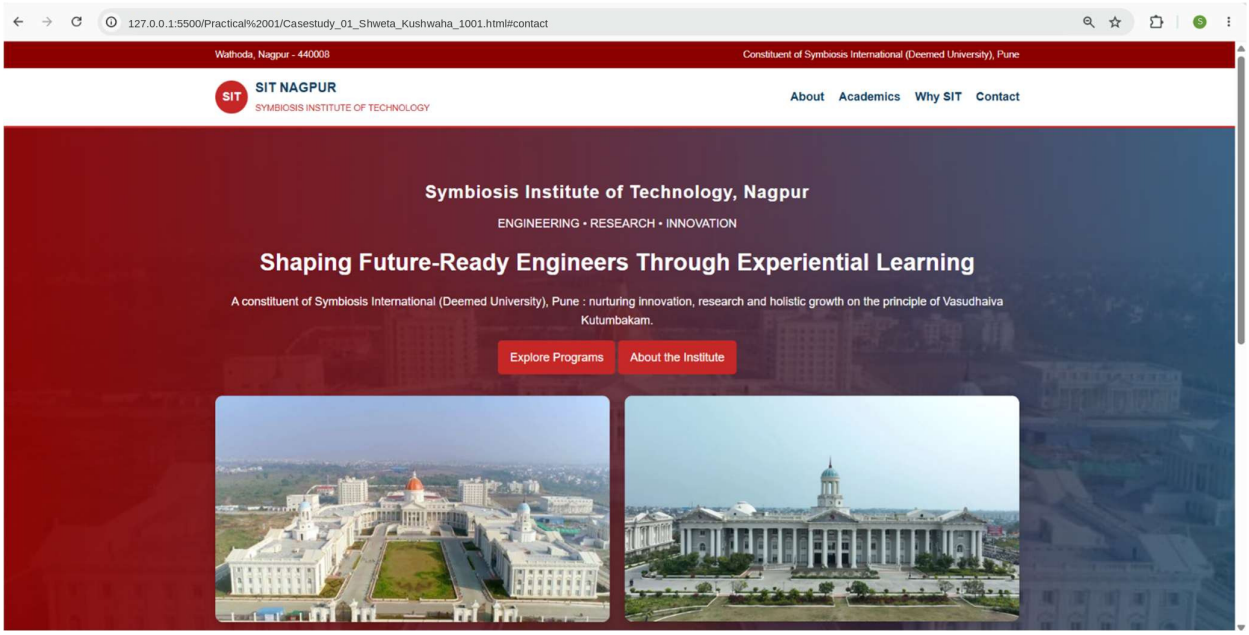
    <script>
        window.alert('Welcome to SIT');

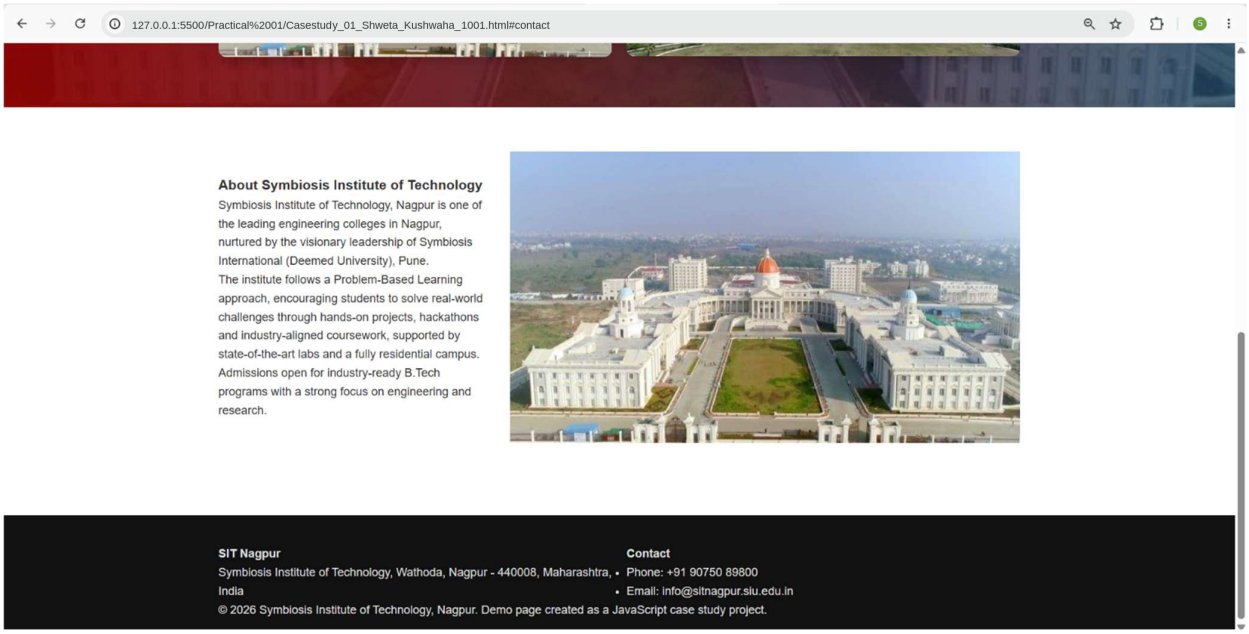
        var toggle = document.getElementById('navToggle');
        var navUl = document.querySelector('nav ul');
```

```
        toggle.addEventListener('click', function () {
            if (navUl.style.display === 'block') {
                navUl.style.display = 'none';
            } else {
                navUl.style.display = 'block';
            }
        });
    </script>

</body>
</html>
```

7. Output of the case study :





8. Result/Conclusion

Thus, the different ways of integrating JavaScript into a webpage: inline, internal, and external, along with console methods (console.log, console.table, console.trace) were successfully demonstrated. A case study webpage modeled on the SIT Nagpur institute home page was also implemented, using window.alert() to display a welcome message on page load and internal JavaScript (DOM manipulation via addEventListener) to build a responsive navigation menu toggle, thereby demonstrating practical, real-world usage of JavaScript for dynamic and interactive web pages.